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Innovative Monitoring Technologies for Well Control Through Sensor Integration and Edge Computing

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Abstract

Effective monitoring of well performance is vital for optimizing production and operational management in the oil and gas industry. This study presents advanced monitoring technologies integrating digital twins, multisensor networks, and edge computing to enhance real-time data processing capabilities. Laboratory experiments verified the effectiveness of clamp-on vibration sensors over piezoelectric accelerometers, particularly in high-gas-factor scenarios where the latter's performance is severely impaired due to bubble-induced signal disruption. A hybrid measurement system integrating vibration, pressure, and temperature sensors was developed, allowing comprehensive and accurate real-time monitoring of well conditions, equipment anomalies, and the presence of mechanical impurities.

The incorporation of mobile measurement units integrated with digital twin technology enables significant reduction in reliance on stationary installations, thus minimizing capital expenditures and operational downtime. A sophisticated data analysis methodology utilizing machine learning and spectral analysis of sensor signals was established, enhancing the accuracy of digital twins and facilitating virtual flow metering. Edge computing technologies enable rapid data processing at the point of acquisition, significantly reducing data transmission volumes and improving responsiveness to operational changes.

The proposed system offers considerable potential for practical application by providing precise, non-invasive, and adaptable monitoring solutions capable of enhancing decision-making, reducing operational risks, and improving economic efficiency in complex oilfield environments.

Introduction

In the modern oil and gas industry, effective well monitoring is a key element in optimizing production processes and managing operations at oilfields. The accuracy and timeliness of data obtained during monitoring directly influence decision-making, hydrocarbon production optimization, and, consequently, the overall profitability of the company. As production processes become increasingly complex and the need to reduce capital expenditures grows, the issues of control and optimization of production become especially relevant.

This study focuses on developing and implementing advanced well monitoring technologies based on digital twins, integrated clamp-on vibration sensors, and edge computing for real-time data processing. The use of machine learning algorithms for analyzing signals from measurement systems significantly improves the accuracy and reliability of operational process control. The introduction of these innovative approaches not only enhances data quality but also simplifies the decision-making process, which is critically important for the prompt management of well operations.

Particular attention in this study is paid to the development of a hybrid measurement system combining vibration, temperature, and pressure sensors. The use of mobile measurement systems integrated with digital twins enables the replacement of stationary installations, leading to reduced operational costs and increased economic efficiency of oilfields.

Thus, the aim of this study is to create an efficient monitoring system capable of providing high measurement accuracy and a timely response to changes in well operation. The results of the work can form the basis for further improvement of production technologies and operational management in the oil and gas sector, ultimately contributing to increased productivity and reduced operational risks.

Review

Modern monitoring systems, based on the integration of multiple sensors and the use of edge computing, enable real-time flow parameter analysis, detection of production anomalies, and the creation of digital twins to support decision-making. A literature review shows that accumulated research in vibration measurement, signal analysis, and non-invasive flow monitoring methods provides a solid foundation for developing advanced well control systems [1].

Numerous studies confirm the advantages of non-invasive methods, particularly clamp-on sensors. Research focusing on the use of vibration sensors shows that in high gas content conditions, these devices outperform acoustic methods, as even the smallest gas bubbles significantly distort the passage of acoustic waves [1]. Additionally, the use of piezoelectric accelerometers to measure pipeline vibrations has enabled accurate assessment of liquid flow rates and the detection of dynamic changes in flow regimes [2]. Moreover, recent studies demonstrate that combining data from pressure, temperature, and vibration sensors allows for a more comprehensive understanding of well processes, which is especially important for constructing digital twins and implementing adaptive process management [3].

Spectral analysis methods are actively used to interpret signals obtained from clamp-on sensors. The application of FFT, STFT, and wavelet analysis allows the extraction of characteristic frequency ranges correlating with various flow regimes in the well. These methods not only ensure high measurement accuracy but also enable the use of machine learning algorithms to build predictive models that link signal dynamics with flow parameters [4]. Through the integration of modern signal processing algorithms, it becomes possible to develop adaptive models capable of real-time adjustment of measurement parameters according to changing operating conditions.

In recent years, the oil and gas drilling industry has been actively exploring digital and intelligent approaches for early anomaly detection in wells. One promising direction is the use of digital twins, multisensor networks, and machine learning algorithms executed on local computing nodes (edge devices). A digital twin is a virtual model of a physical object (e.g., the well or drilling rig) containing all engineering information about the system and replicating its behavior in real time. Essentially, the digital twin establishes a digital connection with the real well, allowing for the prediction of its state and deviations from the norm.

The application of edge computing ensures data processing directly at the point of collection, significantly reducing latency and improving decision-making speed. Recent studies have shown that mobile systems equipped with multisensor devices can replace numerous stationary installations, reducing capital costs while maintaining high monitoring accuracy [5]. Furthermore, the use of flexible machine learning

algorithms that adapt to changing conditions allows integrating data from various sensors into a unified digital twin model, which is a promising area for further research [6].

Despite significant progress in non-invasive well monitoring, challenges remain regarding sensor calibration under changing flow conditions, adaptive signal processing, and integration of different data types. Future research should focus on developing universal algorithms for automatic digital twin updates, improving model accuracy using edge computing, and expanding the functionality of multisensor systems. Such advancements will enable more efficient well control, minimize downtime, and increase the economic efficiency of production.

Research methodology

Application of a clamp-on sensor system for production monitoring tasks

Modern well production monitoring tasks require the implementation of intelligent systems capable of continuous data collection, processing, and analysis in real-time without interfering with the technological process. In this context, the clamp-on measurement complex represents an optimal solution that expands monitoring and control capabilities while minimizing costs and avoiding well shutdowns.

The clamp-on measurement complex is a system designed to be installed on the external surface of the pipeline. It includes a set of clamp-on sensors, data collection and processing modules, and data transmission devices. The primary task of the complex is to perform real-time measurement of multiphase flow parameters and equipment condition, followed by the integration of these data into the overall digital infrastructure of the field.

The system addresses the following tasks:

- Monitoring well production parameters: pressure, temperature, fluid-induced vibrations, and flow rate;
- Determining phase composition and flow regime in real-time;
- Detecting equipment anomalies;
- Improving diagnostic accuracy through correlation analysis of multichannel signals from vibration acoustic sensors;
- Enabling simplified data integration into digital twin systems and providing additional information to refine analytical well models.

The complex includes the following main components:

- Pressure sensors are installed at the wellhead, ESP intake, or manifold collector. These sensors measure basic flow parameters: absolute and gauge pressure, and dynamic pressure changes over time. They help assess filtration resistance, locate pressure drop zones, and predict potential deviations from the normal mode.
- Temperature sensors are used to determine the flow temperature at reference points such as the ESP intake, wellhead, and manifold collector. These measurements are necessary to recalculate parameters to local conditions, refine flow regimes, and increase the accuracy of analytical models. Temperature data help increase the number of features in training datasets, identify correlations (e.g., an increase in the gas factor is accompanied by a temperature decrease), and use them in three-phase flow models. They can also be integrated with other parameters: pressure, vibration, phase flow rates, to perform a more accurate system analysis.
- Vibration acoustic sensors are mounted on wellhead equipment elements and other pipeline sections. Their primary function is to record mechanical vibrations and acoustic signals generated by fluid flow. Analyzing the time and frequency structure of these signals allows:

- Recognition of flow regimes (bubble, slug, annular);
- Detection of cavitation processes, gas plugs, water hammer;
- Diagnosis of pump equipment instabilities;
- Classification of current conditions using acoustic features with machine learning methods.
- Piezoelectric elements are designed to provide a wide frequency range to capture both low-frequency and high-frequency noises. This is essential for covering all potential noise sources. High sensitivity of piezoelectric elements is necessary for detecting weak noise signals, allowing for more precise and reliable measurements even at minimal sound pressure levels. In the integrated sensor set used for monitoring well operations, piezoelectric accelerometers are presented in two configurations: combined and separate. In the combined variant, two receivers (two piezo plates) oriented toward passive acoustics are placed opposite each other on the pipeline walls. In the separate variant, the receivers, also oriented towards passive acoustics, are installed on one side of the pipeline at the bend: one before the bend and one after.
- Data collection and processing modules include analog-to-digital converters, computational units, and built-in preprocessing algorithms (filtering, normalization, signal decomposition). These modules ensure local data aggregation, noise reduction, and data preparation for subsequent analysis.
- Communication and data transmission tools include wireless modules or wired interfaces that transmit information to the upper level (SCADA systems, data centers, cloud analytical platforms). These tools incorporate secure transmission protocols, synchronization with external time sources, and adaptation to unstable communication channels.

The use of a clamp-on measurement complex significantly improves the accuracy and frequency of measurements, eliminates dependence on traditional measurement installations, and extends the range of monitored parameters. Combining several sensor types provides a more comprehensive understanding of ongoing processes, which is especially crucial when operating under complex geological and technical conditions and when dealing with unstable flows.

Experimental setup and laboratory investigations

To test the hypotheses proposed within the framework of the study, a series of experiments was conducted on specialized laboratory setups designed to simulate operational conditions for multiphase gas-liquid mixtures (GLM) within well equipment components. The primary objective of these experiments was to explore the feasibility of integrating multisensor systems and edge computing technologies to enhance well condition monitoring and hydrocarbon production management.

The first set of experiments was performed on a test bench that simulates GLM flow through tubing, casing and wellhead equipment (Fig. 1). The tests were carried out with liquid flow rates ranging from 75 to 150 m³/day. For each liquid flow rate value, gas flow rates were selected to ensure the required volumetric gas content in the flow, ranging from 30% to 90% (by volume). During the experiments, liquid and gas flow rates were recorded using high-precision measuring devices presented in Fig. 1, thus providing reference values for evaluating the accuracy of the developed monitoring systems (1 – tank; 2, 19, 26 – cantilever centrifugal pump; 3, 20, 27 – electric motor; 4, 21, 28 – frequency-controlled drive; 5, 22 – mass flow meter; 6 – solid particle input block; 7 – piston compressor; 8 – gas rheometer flow meter; 9 – gas inlet tubes; 10 – casing column; 11 – pump body; 12 – pipeline valve; 13 – annular valve; 14 – three-way valve; 15 – bypass line with valve; 16 – linear valve; 17 – cyclone separator; 18 – mechanical impurity filter; 23 – separator valve; 24 – gravity separator; 25 – gas meter).

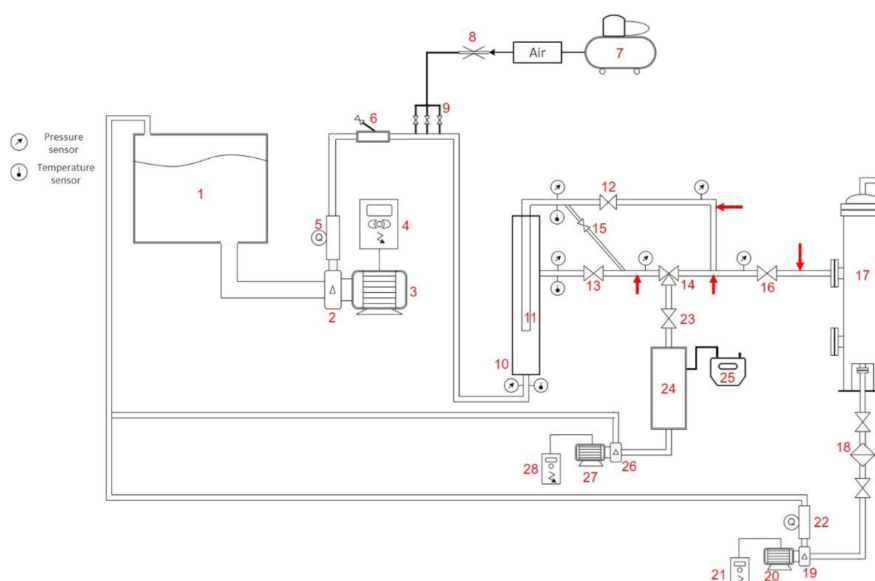


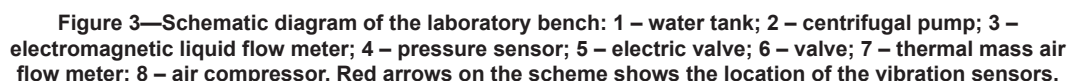
Figure 1—Schematic diagram of the laboratory bench. Red arrows on the scheme shows the location of the vibration sensors.

Additionally, vibration sensors were installed on this setup at locations with the highest flow disturbances, marked on the diagram with red arrows: on the annular line, at the manifold branch, at the tee connecting the annular line and the manifold branch, as well as on the common flow line.



Figure 2—The laboratory bench.

The next series of experiments was conducted on a laboratory setup, the schematic of which is shown in Fig. 3. This bench simulated various GLM flow regimes using a centrifugal pump to control the liquid flow rate and a compressor to supply gas to the system. Clamp-on vibration sensors were installed to analyze flow dynamics and detect different types of pipeline disturbances. The sensors were positioned at characteristic points where the flow trajectory and structure changed: immediately after the shut-off valve and after a 90° pipe bend. These experiments aimed to investigate the sensitivity and reliability of vibration acoustic methods for monitoring well equipment condition and detecting deviations in GLM flow regimes.



The diagram illustrates a vacuum vacuumation system. It features a vacuum tank (1) at the top, which feeds into a vacuum breaker (2). Below the vacuum breaker is a vertical pipe (3) with a check valve (4). This pipe is connected to a main vacuum line (12) that runs horizontally. On the left, this line branches off to a vacuum breaker (9) which serves two storage tanks (7 and 8). Tank 7 has a float valve (10) and a vacuum breaker (11). A separate line (14) also connects to tank 7. On the right, the main vacuum line (12) connects to a vacuum pump (6) via a complex piping system that includes a vacuum breaker (13) and a vacuum breaker (14). The pump (6) is shown with a motor and a vacuum gauge. The entire system is designed to maintain a vacuum in the sewer lines to facilitate the removal of solids and liquids.

Figure 4—Schematic of the SPC test bench: 1 – electric motor, 2 – receiving module, 3 – ESP, 4 – gas separator with axial unloading element at the lower base, 5 – frame, 6 – compressor unit with gas flow measurement function, 7 – tank with a 120 L working mixture, 8 – container with flushing liquid, 9 – tee, 10, 11, 13 – shut-off valves, 12 – working mixture supply unit, 14 – discharge line of the working mixture.

Data Analysis Techniques and Digital Twin Integration

The data analysis stage begins after the preliminary cleaning and normalization of signals obtained from clamp-on vibration sensors and clamp-on measuring complexes. This stage employs time-domain, frequency-domain, and time-frequency analysis methods, enabling the identification of key flow behavior patterns, typical flow regimes, and anomalies indicating potential deviations from the system's normal state.

Frequency analysis of acoustic signals is based on the application of the FFT, which converts time-domain data into a spectral representation. FFT usage highlights the frequency components of signals and quantifies parameters such as the average amplitude within specified frequency ranges and the spectral centroid. It has been shown that the spectral centroid has a high correlation with the liquid phase flow rate, while high-frequency amplitudes correlate with the gas flow rate.

A more in-depth study of vibration acoustic signals is carried out using Mel Frequency Cepstral Coefficients (MFCC), which transform spectral data into a Mel-scale that reflects human auditory perception of frequencies. MFCC provides a compact and informative signal representation, effectively revealing relationships with flow parameters, such as:

- Flow regime (e.g., slug or annular), due to pronounced spectral features;
- Phase composition of the mixture (sensitivity of coefficients to changes in the gas–liquid ratio);
- Gas factor (distinct spectral patterns when the gas content increases).

Additionally, MFCC analysis allows identifying discrete phenomena within the flow, such as water hammer and cavitation, which significantly impact the stability and reliability of equipment operation.

Correlation analysis plays a crucial role in data analysis, aiming to identify and quantify relationships between different physical flow parameters, such as pressure, temperature, vibration, and phase flow rates. This approach enables:

- Assessing the mutual influence of parameters (e.g., the effect of a higher flow rate on the increase in wellhead pressure);

- Building regression models to predict flow rate and phase composition based on indirect features;

- Identifying causal relationships in system behavior;

- Reducing data dimensionality by selecting the most informative features.

For a detailed analysis of transitional processes and unstable regimes, time–frequency analysis (e.g., spectrograms and wavelet transformations) is used to detect transitions between different flow regimes and generate event signatures for subsequent automated classification.

Derivative calculation methods, such as pressure and temperature gradients, are also employed to record dynamic changes in flow conditions.

Integrating the obtained data with digital twins requires accurate temporal synchronization of signals from sensors located at different pipeline points. To solve this problem, time delays (lags) between measurement points are calculated based on stationary hydraulic correlations, taking into account fluid flow characteristics, physicochemical properties of the medium, and pipeline geometry. This significantly improves the accuracy and reliability of digital twins, enhancing modeling quality and real-time control of well operations.

To enhance the accuracy, stability, and automation of data analysis from the clamp–on measurement complex, machine learning algorithms are integrated into the system, including linear regression models and gradient boosting methods. These algorithms dynamically adapt analytical models to changing operating conditions. The models are trained on labeled datasets collected from various operational modes on laboratory stands. Special attention is given to class balance, noise resistance, and model generalization ability. The developed models are deployed on edge devices or cloud infrastructure, providing online analysis and predictive diagnostics.

Comprehensive utilization of machine learning algorithms along with correlation, frequency, and time-frequency analysis methods forms the intelligent foundation of the monitoring system, capable not only of recording current parameters but also of predicting well behavior under changing operational conditions.

Edge computing

Edge computing has become a vital solution for modern well monitoring systems in the oil and gas industry, where large volumes of data are generated from various sensors, such as vibration acoustic sensors, temperature sensors, and pressure sensors. Transmitting raw data directly to centralized processing systems incurs high costs related to network infrastructure and increases the risk of delays in decision-making, especially in operational environments where timely responses are critical.

Edge computing technology addresses these challenges by performing initial data processing and analysis directly near the data source, close to the sensors at the equipment site. This approach significantly reduces the volume of data transmitted to central processing centers since only preprocessed and aggregated parameters are forwarded. By minimizing data transmission, edge computing improves the responsiveness of monitoring systems and ensures timely decision-making.



Figure 5—Control panel with an industrial computer installed.

An edge computer, specifically designed for harsh conditions typical of oil and gas production sites, performs real-time data processing. This includes spectral analysis of signals, calculation of spectral centroids, and average amplitudes of vibration acoustic signals over a wide frequency range. These processed data points are then used to calculate key equipment condition indicators and liquid and gas phase flow parameters.

After processing and aggregation, the software on the edge computer sends the final aggregated metrics (such as flow parameters and technical condition indices) to a programmable logic controller (PLC), which integrates the data with upper-level management systems (SCADA). This structure minimizes information transfer delays, reduces the load on existing data transmission networks, and significantly accelerates the decision-making process, ensuring high reliability and efficiency in equipment monitoring.

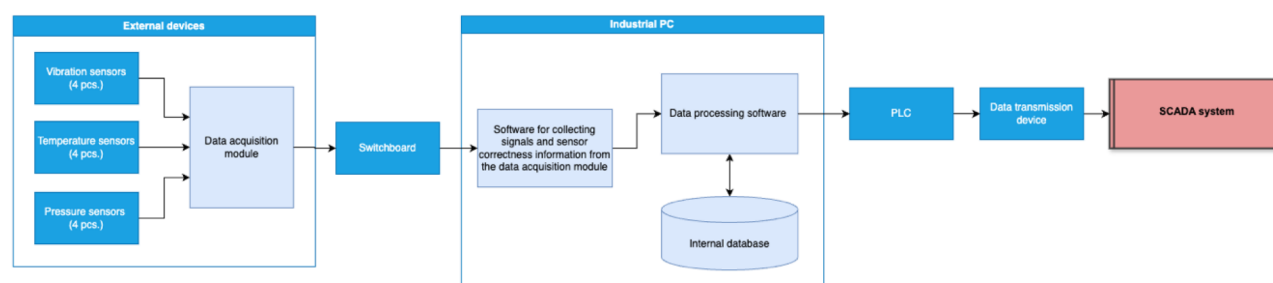


Figure 6—Scheme for receiving, processing and transmitting data.

The implementation of edge computing technology significantly simplifies the overall architecture of the monitoring information system, reducing operational and maintenance costs. Utilizing edge computing ensures accurate and timely information, which is essential for the efficient management of oil and gas facilities and for making prompt managerial decisions.

Experimental results

As part of the conducted laboratory research, an analytical model based on the processing of signals obtained from vibration acoustic sensors was tested. The primary objective was to predict the liquid flow rate on a laboratory stand. The comparison between actual and predicted values confirmed the high accuracy of the developed model, as evidenced by the low value of the symmetric mean absolute percentage error (SMAPE = 8.92%).

Figure 7 below shows a comparison between the actual liquid flow rate recorded by traditional measurement systems (blue line) and the flow rate values predicted by the model (red dashed line). The graph clearly illustrates that the model effectively tracks dynamic changes in the liquid flow rate, including sharp spikes, transitional modes, and stabilization periods. This demonstrates the high sensitivity of the developed approach, based on the processing of vibration acoustic signals, and justifies the use of comprehensive analysis methods, such as cross-correlation and time-frequency analysis, within the monitoring system framework.

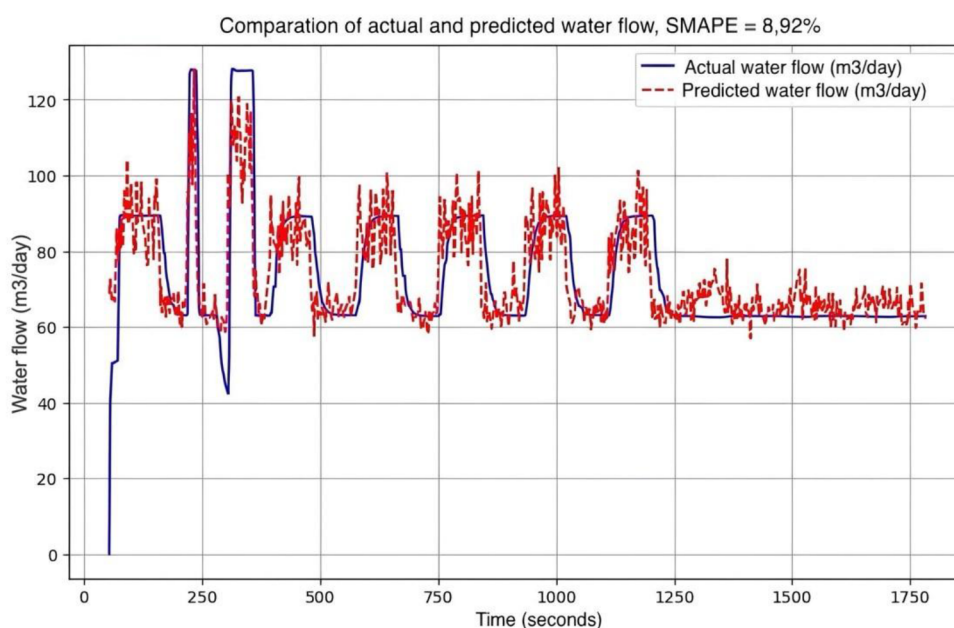


Figure 7—Comparative graphics of actual and predicted by analytical model based on vibration sensors water flow.

Additionally, laboratory experiments were carried out to evaluate the possibility of detecting the presence and passage of suspended solid particles (e.g., sand) in the flow using clamp-on piezoelectric accelerometers. The obtained data demonstrated that the spectral characteristics of the signal significantly vary depending on the gas content in the gas-liquid mixture (GLM). Specifically, at low gas volume fractions (0-0.4), a dominant noise frequency component up to 2500 Hz was observed, reflecting the typical acoustic profile of GLM movement with low gas content (Figure 8). As the gas fraction increased (above 0.4), a substantial broadening of the noise spectrum and an increase in intensity in the region above 2500 Hz were observed, corresponding to the passage of air plugs and changes in the flow structure (Figure 9).

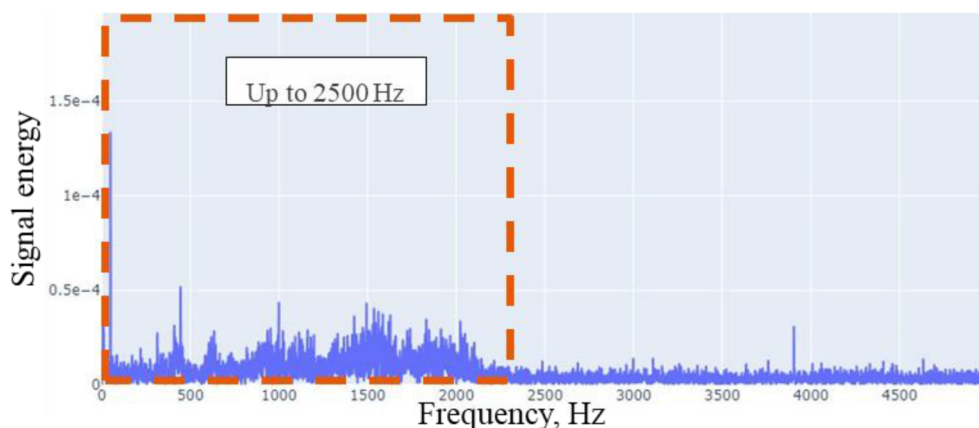


Figure 8—Spectrum for the fraction of gas in the flow up to 0.4.

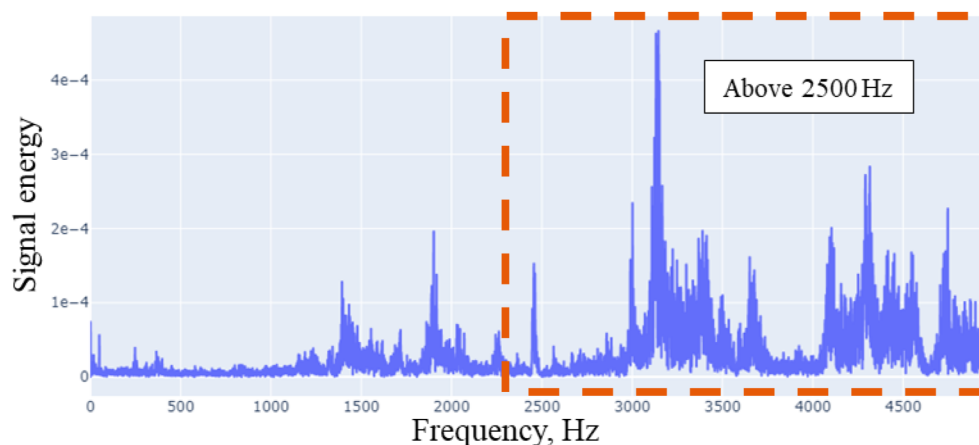


Figure 9—Spectrum for the fraction of gas in the flow is more than 0.4.

However, a further increase in gas content (above 0.7-0.8) was accompanied by a significant deterioration in the transmission of acoustic waves through the flow due to high dispersion and damping of the acoustic signal. This indicates the fundamental limitations of using high-sensitivity piezoelectric accelerometers under field conditions, where the gas content in the flow can reach significantly higher values. Therefore, it is advisable to combine acoustic methods with other types of sensors and analytical approaches (e.g., vibration acoustic) to ensure stability and accuracy in monitoring multiphase flow over a wide range of operating conditions.

Flow through the annular space

During the course of the research, a series of experiments were conducted to identify various stable and transient operating modes of the well, including those involving flow through the annular space. Special

attention was given to analyzing the behavior of gas-liquid mixtures passing through the annulus, with the aim of identifying perturbations that could serve as indicators of changes in the well's operating mode.

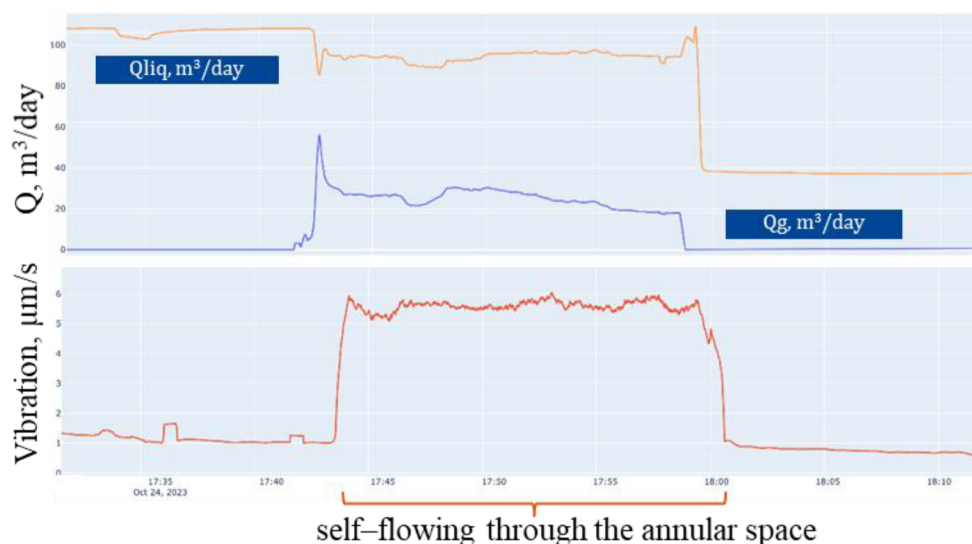


Figure 10—Analysis of self-flowing through the annular space using vibration sensors.

Objective data was collected using sensors installed at key points within the system. An analysis of the collected data revealed a clear correlation between the dynamics of the indicators, which enabled precise identification of moments when the operating modes changed.

The results of the observations and experiments lead to the conclusion that the overhead measurement system is highly effective for monitoring transient well operating modes. These findings confirm that such systems can be used for the real-time recording and analysis of changes during well operations, significantly enhancing the ability to monitor their technical condition and improving the accuracy of predicting potential deviations in operating modes. Therefore, the proposed methodology is recommended for widespread application in well operations, providing improved safety and operational efficiency.

Suspended particle concentration

The analysis of suspended particle concentration (SPC) in multiphase flows is a critical aspect of well monitoring. The presented graph illustrates the relationship between the introduction of SPC into the system and the displacement of piezoelectric sensors. During the initial phase, a significant slug discharge occurs, indicated by a rapid increase in displacement as SPC concentration reaches 1800 g/l. As the experiment continues, the SPC concentration stabilizes at approximately 720 g/l, while the displacement remains elevated, reflecting the continued presence of suspended particles.

In the final phase, a flushing process is conducted to reduce the SPC concentration, resulting in a decrease in sensor displacement. This demonstrates the dynamic response of piezoelectric sensors to changes in suspended particle content, indicating their sensitivity to sudden surges and subsequent stabilization of the multiphase flow. The results emphasize the need for integrating acoustic measurement with other sensor types to maintain accurate monitoring under varying flow conditions.

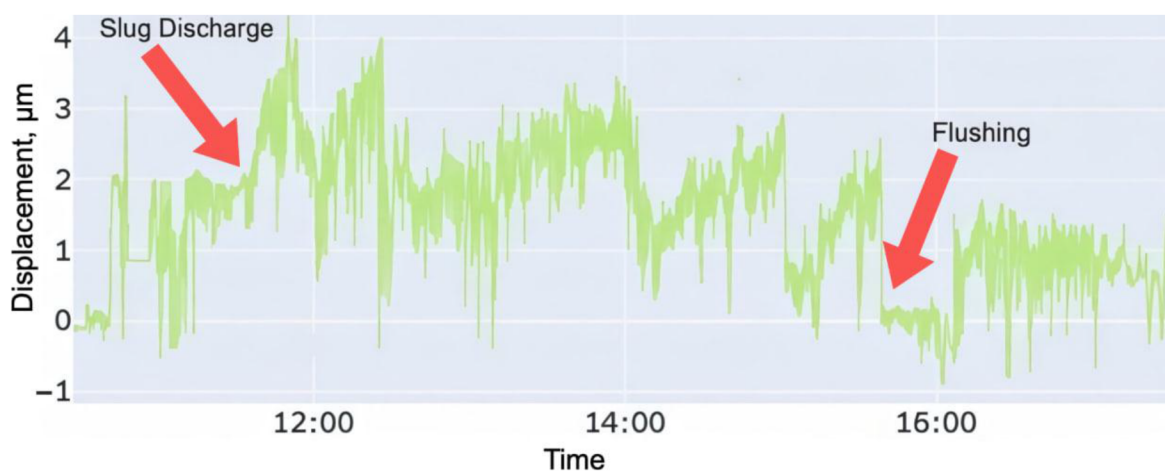


Figure 11—Displacement dynamics of piezoelectric sensors during slug discharge and flushing in multiphase flow with suspended particles.

Discussion

The developed well monitoring system based on a clamp-on measurement complex with vibration acoustic sensors for liquid flow rate estimation offers significant advantages for real-time, non-invasive monitoring without interrupting the technological process. The system effectively tracks changes in production regimes, demonstrating high sensitivity to flow fluctuations, including spikes and stabilization periods, through cross-correlation analysis of vibration acoustic signals.

Advantages of the proposed system include:

- Non-invasive and flexible installation: clamp-on sensors are mounted on the external surface of the pipeline, eliminating the need for intrusive installation and process downtime, thereby reducing operational risks and installation costs.
- High flow sensitivity: the system accurately detects rapid changes in flow rate, leveraging advanced signal processing methods.
- Multifunctionality: the integrated complex simultaneously monitors pressure, temperature, and vibration, providing a comprehensive assessment of well conditions.
- Intelligent data processing: utilizing machine learning algorithms enables not only current parameter monitoring but also predictive modeling based on production dynamics.

Challenges and limitations:

- Sensitivity to external factors: external vibrations and noise can distort the signal, necessitating advanced filtering and calibration techniques.
- Installation quality dependence: proper mounting of vibration acoustic sensors is critical for accurate measurement.
- Limited applicability in complex environments: accuracy may be affected in heterogeneous pipelines or with significant temperature variations.

While these challenges exist, they can be addressed through advanced data preprocessing methods, including adaptive filtering, spectral analysis, and signal decomposition. These techniques help mitigate noise and external disturbances, improving the quality of input data and the accuracy of forecasts and diagnostics.

Mobile clamp-on measurement systems, with their flexibility, can be effectively integrated into real-time production control at oil fields. Their implementation enables rapid identification of unstable regimes, monitoring of pump efficiency, and real-time well condition analysis. Furthermore, such systems facilitate

the creation of digital well twins, allowing both current state monitoring and predictive modeling for production optimization.

Thus, the proposed system demonstrates substantial potential for the oil and gas industry, particularly under complex production conditions where traditional monitoring methods may lack precision or timeliness.

Conclusion

The study developed and experimentally validated the concept of advanced well monitoring complex based on digital twins, multisensor systems, and edge computing. Laboratory tests on a fountain well model with a gas-liquid mixture demonstrated that the combination of vibration, acoustic, hydraulic, and thermal sensors reliably and promptly detects changes in well operation modes. Comparative sensor analysis revealed that piezoelectric transducers possess high sensitivity, while vibration sensors provide robustness, making the use of vibration sensors the optimal solution.

The developed architecture includes a local computational module implementing machine learning algorithms and a well digital twin, enabling intelligent data processing directly on-site. The use of edge computing ensures instant response to anomalies without data transmission delays, while the mobile system design reduces capital expenditures and simplifies deployment.

The key outcome of the study is a significant improvement in monitoring speed and reliability: the multisensor system with a digital twin can detect well deviations at the earliest stage with minimal false alarms. This confirms the potential of the proposed approach for practical application in oilfield facilities. Implementing such systems in the oil and gas industry will significantly increase well operation efficiency. Moreover, finer and automated control opens up the possibility of optimizing production modes.

In conclusion, the integration of digital twin technologies, AI analytics with modern sensors, and edge computing devices represents an effective solution for intelligent well monitoring. Future research in this direction may involve field trials of the system on operational wells, scaling the data collection infrastructure, and expanding the functionality of the digital twin. Nevertheless, the results obtained already indicate that the proposed technology is viable and has great potential for improving the efficiency and safety of oil production in increasingly complex field developments.

Nomenclature

FFT	– Fast Fourier Transform
STFT	– Short-Time Fourier Transform
ESP	– Electric Submersible Pump
SCADA	– Supervisory Control And Data Acquisition
GLM	– Gas-Liquid Mixture
MFCC	– Mel Frequency Cepstral Coefficients
PLC	– Programmable Logic Controller
SMAPE	– Symmetric Mean Absolute Percentage Error
AI	– Artificial Intelligence
SPC	– Suspended Particle Concentration

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